Revolutionizing Healthcare Management: Architecture of a Web-based Medical Triage Service

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***Abstract*—During the COVID-19 pandemic, the traditional emergency healthcare systems faced unprecedented strain due to the sharp rise in demands for urgent care, scarcity of resources, and increased risks of people getting infected while waiting at the emergency care facility. We present Triage-Bot, an online medical triage provisioning service, that can revolutionize emergency care by decreasing the load on emergency departments (ED), reducing healthcare expenses, and improving the quality of care. Empowered by artificial intelligence and natural language processing, the Triage-Bot service assesses and prioritizes patients' needs based on symptoms, medical history, and perceived conditions from multimodal video, audio, and text data captured during patients' interactions. The captured summarized information with a severity ranking is sent to a human expert to suggest the next action on the user's part. The diverse data types used by the Triage-Bot in communication, authentication, data collection, storage, and analytics requires a robust and scalable system architecture for online service provisioning. In this paper, we specifically focus on the system design and architecture of the Triage-Bot for emergency healthcare settings. With integrated electronic medical records (EMR) and online platforms, the bot fosters collaboration among healthcare professionals and enables swift and informed decision-making even in the face of crises. By partially automating and offering a hybrid triage process, the Triage-Bot improves resource allocation, reduces healthcare management costs for emergency care, minimizes patient waiting times, and improves wellbeing. To address the complexities and demands of healthcare data management, our proposed system incorporates MongoDB database for flexibility, scalability, and versatility in supporting different types of data. Additionally, we implement a data linking and analytics pipeline utilizing a data Lakehouse system to effectively ingest, manage, process, and generate knowledge from heterogeneous data sources.**

***Index Terms*—Database, Triage bot, Healthcare Management, Data Lakehouse.**

I. Introduction

The critical landscape of emergency medical services exacerbated by the COVID-19 pandemic [1], underscored the need for an accessible, reliable, and timely urgent care service, as every passing moment significantly impacts patient outcomes [2][3]. Quick and accurate assessments during the triage process are crucial for prioritizing care and resource allocation [4]. Manual triage systems can be leveraged by complementary automated triage services [5][17], which can prioritize patients and allow patients with less urgent problems to be treated at home to help reduce the load on the emergency department (ED). A controlled automated triage service can allow patients with compromised mobility to get urgent and reliable care from a home environment, thus reducing healthcare costs and improving the well-being of both patients and caregivers. Future healthcare management systems can greatly benefit from the seamless collection of rich multimodal data comprising patients' reports, remotely measured vital signs, and correlated information from patients' historical electronic medical records (EMR) [6]. The data would enable the development of advanced predictive and prescriptive analytics models for disease prediction and prevention [8].

There has been a growing recognition of the need for innovative solutions to enhance emergency healthcare systems [19]. Remote healthcare systems have emerged based on advanced artificial intelligence (AI) and machine learning techniques [20][21]. This paper introduces a novel Triage-Bot system design and architecture for provisioning online triage services at the EDs. We developed a variety of novel techniques and integrated these using the proposed backbone infrastructure to enable the following features.

1) *Security*: Allow new users to register and old users to login [9] to a secure service platform and perform secure multiparty computation [10] to ensure data security.
2) *Multimodal User Interaction*: Provide an accessible online user interface for multimodal audio, video, chat conversations, and other user interactions [11].
3) *Vital Sign Measurement*: Enable remote measurement of vital signs with users' approvals from face or finger videos [12][27].
4) *Emotion Recognition*: Detect emotions from video and text data to enable empathy and compassion in conversation and to improve criticality assessment [13].
5) *Dialogue Generation*: Generate goal-oriented dialogue to collect necessary information for disease diagnosis and criticality assessment [10] [14].
6) *Consult EMR*: Extract patients' historical health data from an EMR system [15].
7) *Disease Prediction*: Predict a diagnosis based on the acquired information [15].
8) *Severity Assessment*: Categorize patients' severity using

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the CTAS scale [7] based on the collected data.

Therefore, we made a careful selection of implementation and deployment tools to build the pilot prototype of our proposed Triage-Bot service. In this paper, we present the architecture of our Triage-Bot service that supports all the above functionalities. Furthermore, it can ingest, link, and analyze large multimodal data from diverse sources.

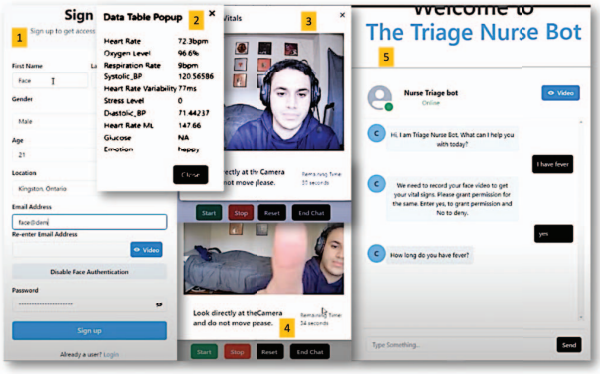


Fig. 1. From left to right: (1) Sign-up page for user authentication; (2) Pop-up table showcasing vital signs of the patient; (3) and (4) Video screens for capturing patient's face or finger videos; (5) TriageBot with patient conversation

A. Contributions

The key contributions presented in this paper are as follows.

- We present the architecture of a multifunctional web-based Triage-Bot service. The architecture is scalable, extensible, and validated using the MIMIC III dataset, a rich repository of real-world critical care data, along with other data acquired from the user interfaces. We use MongoDB, a flexible and scalable database solution, to effectively manage the diverse array of data types commonly encountered in healthcare records.
- We integrate MongoDB with a data Lakehouse system that can ingest large batch and streaming data and enable advanced data analytics by linking heterogeneous data from multiple sources. The analytics capabilities can in turn allow collaborative decision-making by multiple stakeholders as needed.

The rest of the paper is organized as follows. Section II provides a description of the related work. The proposed methodology is explained in Section III. In Section IV, we discuss the implementation and infrastructure design of our proposed Triage-bot system. we then validate the storage infrastructure using predefined queries in section V. Section VI presents the results. In Section VII, we discuss the results of our experiments. Finally, Section VIII concludes the paper by outlining potential avenues for future research.

II. Literature Review

In this section, we present a literature review on chatbots and relevant data management infrastructures.

A. Chatbots

Demasi et al. [17] propose the Crisisbot, a chatbot aimed at training human helpline counselors on diverse scenarios using a low-risk practice environment. The chatbot is designed to emulate numerous visitor stories to provide multi-task training, and incorporates counselling techniques, pre-trained generative models, utterance segmentation, and sub-utterance selection to generate persona-relevant responses based on sample transcripts. The framework was evaluated by both inexperienced crowd workers and seasoned helpline counselors. Although the chatbot significantly enhances response diversity and persona relevance, there remains a need to strengthen conversational coherence. These findings underscore the importance of involving target users in system development and evaluation and highlight the potential applicability of the framework across various contexts.

Nißen et al. [18] explore the impact of different chatbot personas and social roles on the dynamics between clients and chatbots, as well as user intents. The researchers report that user demographics, such as age and gender, influenced perceptions of interactions between customers and chatbots. Younger users show a preference for expert and institutional roles, while older users favored peer and dialogical self-roles. Providing customers with the option to choose a chatbot persona further enhanced human perceptions. This study underscored the importance of tailoring chatbot personas based on user demographics to enhance interactions and engagement.

Angga et al. [19] introduce a chatbot featuring voice interaction and a 3D avatar aimed at creating a more natural and engaging conversational experience. The design allow users to interact with chatbots in a more immersive and human-like manner by integrating speech recognition, chatbot API, text-to-speech, and a 3D avatar with facial expressions, potentially enhancing user satisfaction and the effectiveness of chatbot-based services.

Zhan et al. [20] introduce an AI chatbot system called StoryBuddy, designed to aid parents in storytelling sessions with their children by prompting inquiries about story narratives. This approach addresses the challenges parents face in finding time for storytelling and meaningful questioning. While existing methods often rely on handwritten questions and a limited selection of books, StoryBuddy leverages advancements in natural language processing and comprehension to offer a more engaging storytelling experience tailored to the parent's educational objectives and the child's developmental stage. Additionally, the system monitors a child's progress and provides insights on parent-AI collaboration, user and environment adjustments, and the enhancement of parent-child relationships through technology.

Bohm et al. [21] present a user-centered chatbot design, which include a case study on the implementation of hiring chatbots in Germany. The authors focused on identifying and analyzing user intentions defined as the goals or information requests that users aim to fulfill through text input. To train AI models to accurately match user input with predetermined user intents, the article highlights the significance of training questions. Additionally, the study presents the process of user-

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centered intent identification and methods for intent analysis and consolidation. It also emphasized conversational design and the selection of dialogue content to have a logical flow. The study provides an examination of the fundamental components involved in chatbot development using the Rasa framework.

Sapna et al. [22] introduce a system named "Recommendence and Fashionsence", which utilizes AI to personalize online shopping experiences. The system features Athena, a natural language chatbot, which engages users and provides customized fashion recommendations. These recommendations were generated by two main machine learning components: Fashionsence, a deep learning model trained on online trends and stylist data to understand fashion preferences, and Recommendence, a combination of deep learning and graph-based collaborative filtering recommenders. By leveraging an online platform, the system aimed to replicate the personalized shopping experience found in brick-and-mortar fashion stores.

B. Data Management

Megdiche et al. [22] propose Data Lake, a Big Data analysis solution, for storage, processing, and transforming big unstructured data. Data transformation within a Data Lake entails considerable effort when ingesting diverse data types (structured, semi-structured, unstructured), applying myriads of transformation operators, and involving different categories of users, including data scientists and analysts. The objective of the proposed Data Lake is to aggregate diverse medical data from a complex information system comprising over 100 large or small datasets for future analysis.

Ayesha et al. [27] introduce a front-end web interface designed to enable users to access the system, request camera access, and initiate video capture. The captured video is transmitted to the backend for vital sign computation, alongside optional user data collection such as age, skin color, and other pertinent information. The back-end system uses a Python application hosted on the server, where videos received from the frontend subsystem are stored, processed, and analyzed to extract raw PPG signals. For streamlined installation, ease of use, and portability, the researchers opted for the SQLite database as the repository on the back-end server to store experimental data. This database comprises a single relational table, which records timestamped data for each user session, including the video file name, calculated vitals, and user-entered measured vitals.

Harby et al. delineate the desired attributes of a data Lakehouse (LH) system which include accessibility, flexibility, and affordability [28][29]. The LH architecture aims to facilitate data ingestion, preprocessing, knowledge extraction, linking, and cost-effectiveness by implementing knowledge graphs and incorporating pertinent information about linked data to enhance metadata management and analytics capabilities. Emphasis is placed on ensuring the accuracy and integrity of the data as it passed through a rigorous transformation and summarization pipeline prior to being stored or loaded into memory. Recommendations are made for data storage solutions such as Snowflake or Amazon S3, and for data ingestion tools such as Egeria, Apache Flink, and Apache NiFi. To address challenges such as data swamps and the need for faster execution of OLAP queries, the authors suggest the development of an advanced and intelligent data ingestion pipeline with knowledge extraction and linking capabilities, other suggestions include an implementation of data profiling methods to identify significant and anomalous data for storage optimization and enhancing data linking capabilities.

Our research extensively examines the existing literature on chatbot design architecture and data management systems to obtain valuable insights on effective design principles. By leveraging this knowledge, we developed our Triage-Bot with a focusing on a user-friendly interface, efficient data management, and data-driven insights for decision-making.

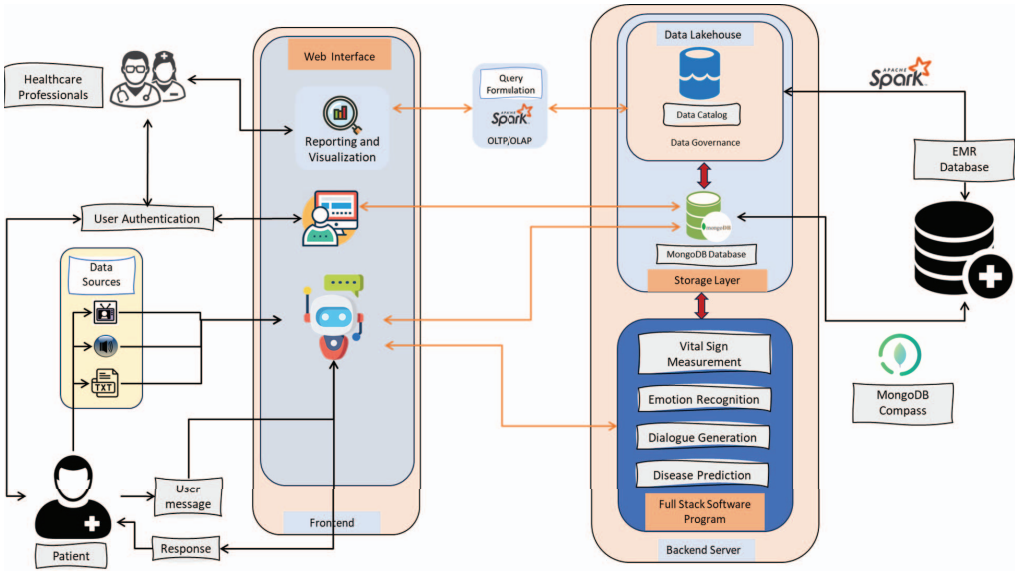


Fig. 2. Proposed Triage-Bot system architecture showing data flow and the system components.

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TABLE I
DESCRIPTIONS OF EACH ENTITY IN THE MONGODB TO SUPPORT THE TRIAGE-BOT FUNCTIONALITY

Entity	Purpose
`user_info`	Stores information about registered patients.
`model_results`	Records the results of our model analysis on patient videos/audios.
`patient_transcript`	Captures conversational transcripts of patients interacting with the chatbot.
`patient_audio_logs`	Logs audio data associated with a patient's interactions.
`patient_audio_logs.files`	Uses GridFS to allow storage of unstructured data. Stores file-level metadata.
`patient_audio_logs.chunks`	Uses GridFS to allow storage of unstructured data. Breaks down audio into manageable chunks.
`patient_video_logs`	Logs video data associated with a patient's interactions.
`patient_video_logs.files`	Uses GridFS to allow storage of unstructured data. Stores file-level metadata.
`patient_video_logs.chunks`	Uses GridFS to allow storage of unstructured data. Breaks down the video into manageable chunks.
`audio_dataset`	Stores audio data for various uses. Used for testing purposes, accessing sample datasets.
`video_dataset`	Holds video data for different purposes.Used for testing purposes, accessing sample datasets.

III. METHODOLOGY

We propose and implement a pilot prototype of Triage-Bot, a web-based medical triage service, to leverage healthcare management systems with managing workload at the emergency departments (ED). Users can register with different roles such as patients, doctors, and caregivers to seek help or provide care with easy access to patients' data. This paper specifically presents a scalable and distributed Web-based infrastructure of the robot by integrating multiple components such as a frontend web-based interface supporting audio/video capabilities, a scalable NoSQL database such as the MongoDB [30], a data Lakehouse (LH) [29] such as the Delta LH, and a backend server deploying a stack of software components to support the various functionalities as illustrated in Fig. 2.

A. Overview of Functional Features

Section I lists the key functional features demonstrated by the Triage-Bot. It allows users to register in different roles as patients, doctors, and caregivers to use the system. New patients can register or log in to converse with the bots, explain their health issues, get their vital signs measured from the face or finger videos captured using the camera on the mobile or desktop device, and get advice about the next step after being ranked based on the criticality of their health situation. Implementation details about the different functional features are out of the scope of this paper. However, we provide an overview of the functional features as they are supported by the proposed infrastructure. A combined screenshot of the various interfaces is shown in Fig. 1. We only discuss the infrastructure components and their operations in this work and validate the efficiency of the data storage and analytics functionality through an experimental study.

The signup page offers password-based or face authentication as shown in Fig.1, acting as the initial interaction point, while the login page provides secure access options. Consequently, the system needs to have a data storage system for facial features and user information. We use MongoDB to store the user information.

Upon login, users engage with the Chatbot using text and voice inputs, granting permission to use face or thumb videos for recording vital signs. Emotion is also detected from video and text interactions. These data including the measured vital signs and emotions are stored in the backend MongoDB and used to rank patients based on the criticality of the health situation. For ranking, EMR data is also accessed to consider comorbidity and existing health conditions for more accurate ranking. Different software components are invoked for the above functionality namely vital sign measurement, emotion detection, dialogue generation, disease diagnosis, and patient ranking. These software components are executed on the backend server as shown in Fig. 2. Selected results are sent to the frontend such as the vital signs, generated next dialogue, and the final assessment/advice to be displayed on the user interface.

B. Data Storage Framework

The above functionalities require a robust storage and distributed computing infrastructure. The implemented research prototype of the Triage-Bot ingests and stores multimodal heterogeneous audio, video, numeric, structured, and unstructured text data from both the frontend interface and other relevant data sources such as the EMR data. It can also link to other remote data sources and either ingest such as the hospital data. Optionally, the data sources can be remotely accessed by the Triage-Bot to retrieve only the necessary data. In our design, we include an EMR database and a data lakehouse in the backend to demonstrate the functional features as shown in Fig.1, and the data query and analytics features as explained in the next section. The EMR data is consulted to generate a severity ranking for the patients based on the Canadian Triage and Acuity Scale (CTAS) [7]. CTAS is a standardized tool used in emergency medicine to prioritize patient care based on the severity of their condition.

C. User Data Management

We considered different use cases for multimodal data access and operations in the Triage-Bot and accordingly, we defined data tables to store the required structured and unstructured multimodal data. Table I provides a description of each table aligned with the design criteria of our Triage-Bot system. We designed the entity-relationship diagrams with the various data components as shown in Fig. 3. Each user can have the vital signs measured multiple times in different interactions with the bot, have multiple conversation transcripts, and audio/video logs. Therefore, the following tables demonstrate one-to-many relationships and this design allows comprehensive tracking

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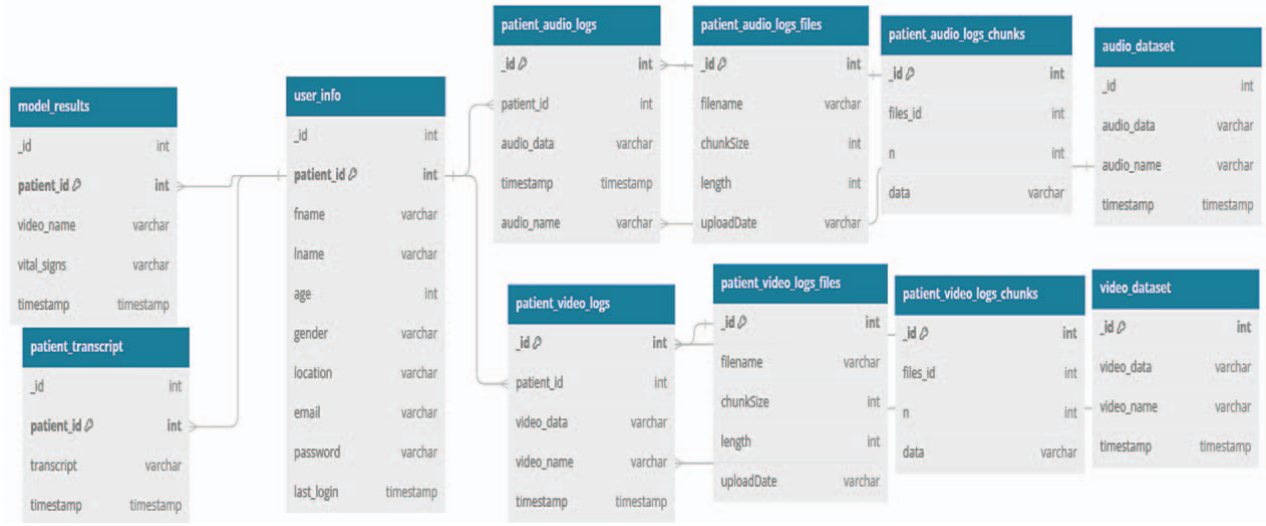


Fig. 3. Entity-Relationship diagram of the database design to support the functionality of the proposed triage bot system.

of user activities and multimedia interactions, enhancing our understanding of user engagement and facilitating personalized care: `user_info` and `model_results`, `user_info` and `patient_transcript`, and `user_info` and both `patient_audio_logs` and `patient_video_logs`.

Furthermore, we employ one-to-one relationships between `patient_audio_logs` and `audio_dataset`, as well as `patient_video_logs` and `video_dataset`. These relationships establish a direct link between the recorded audio/video data and the gridFS storage system, ensuring access to specific audio and video recordings. Through these relationship structures, our data management system optimizes data organization and accessibility, supporting efficient analysis and decision-making in healthcare settings.

IV. Implementation

In this section, we describe the software and storage systems used in the service provisioning infrastructure, including the database design and datasets used in the experiments.

A. Frontend Service Interface Design

In developing the user interface for the services, we strategically selected frameworks and libraries to ensure smooth deployment and compatibility with different devices. We used Next.js and React.js for implementing the frontend interfaces leveraging their modern JavaScript capabilities for easy web deployment. To streamline design and enhance aesthetics, we chose Chakra UI, a robust React-based UI toolkit known for its customizable components and adherence to design principles promoting consistency and accessibility. A distributed modular design approach is used to link the frontend activities to different software modules deployed on the backend Flask server. These software modules process the request and generate appropriate responses either to be stored in the backend database or to send back to the frontend to be displayed on the user interface. Asynchronous execution of the distinct software modules is implemented in the design when possible (no dependency among operations) to provide a seamless communication experience to the user with minimum wait time. For example, when computing the vital signs, the conversation continues with the user and a popup screen opens as shown in Fig. 1 when results are ready to display to the user.

B. Backend Data Management

1) Storage Systems: We use MongoDB [30] to implement the Triage-Bot prototype as it can store multimodal unstructured data, which is critical for storing diverse types of data in a medical voice-video bot. Additionally, MongoDB is horizontally scalable and its flexible data model allows easy schema modifications. This is particularly beneficial as data types and schema can change when new functionalities are added to the bot. MongoDB's built-in replication and sharding capabilities provide high availability and fault tolerance, ensuring data is always available and safe. Its native query language and powerful indexing capabilities allow for efficient and speedy retrieval of data. MongoDB is used to store the user data from bot interactions and implement the EMR system for experimentation.

To demonstrate data linking and analytics we use Delta Lake [31], which is a storage layer in the Databricks Lakehouse platform. It use s a file-based transaction log for atomicity, consistency, isolation, durability (ACID) transactions and metadata handling. It allows streaming and batch operations on structured data. Delta Lake is the default storage format for all operations in Azure Databricks. The protocol is open source and has well-defined SQL commands and a transaction log protocol. We demonstrate a use case where the Delta LH is used to ingest and store data from external data sources such as a hospital database and execute analytical queries on the stored data.

2) Data Processing Tools: MongoDB Compass is a user-friendly Graphical User Interface (GUI) to facilitate visual

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exploration and manipulation of data such as view, query, and analyze data, create databases and collections, and administer tasks like index management and user authentication. It eliminates the need to remember and use intricate commands.

Apache Spark [32] is a powerful open-source distributed computing system for processing large-scale data. It can clean, extract, transform, and ingest real-time data from variety of sources. Spark supports data serialization and deserialization in various formats and can load data into target data stores. It can also check data quality and optimize data storage using partitioning and bucketing techniques. Additionally, Spark integrates well with the components of other big data ecosystems.

3) EMR and Hospital Data Management: We use the Medical Information Mart for Intensive Care (MIMIC-III) [16] dataset to simulate EMR and the hospital data sources. Housing comprehensive health data from over 40,000 critical care patients admitted to the Beth Israel Deaconess Medical Center from 2001 to 2012, MIMIC-III provides a diverse array of patient cases, covering various medical conditions and treatment outcomes. With detailed clinical data spanning demographics, vital signs, medications, and more, MIMIC-III offers a view of patient health journeys. Moreover, the dataset includes free-text clinical notes, enabling the extraction of nuanced patient narratives essential for an online service provisioning triage bot decision-making.

Although we use the same MIMIC III data to simulate both the patient's EMR and hospital data, they have different data structures. The medical data in both sources include diverse patient demography, medical histories, vital signs, and treatment records. Some of the patient IDs exist in both the simulated EMR database and the LH to demonstrate data linking and analytics for ranking patients based on severity.

V. Validation of the Storage Infrastructure

Healthcare professionals can retrieve desired information from the Triage-Bot by linking multiple data sources using Delta LH for disease diagnosis, hospital resource allocation, and generating public health statistics using the Triage-Bot interface. We validate the storage and query functionality of our Triage-Bot infrastructure using 12 predefined OLTP (Q1-Q7) and OLAP (Q8-Q12) queries. We link two or more data tables to demonstrate the advanced decision support capabilities of the proposed Triage-Bot infrastructure. Table III presents the Queries (Q1-Q12) which illustrate data retrieval and management tasks performed by healthcare professionals. We selected various tables from the MIMIC III dataset to be ingested and then queried as shown in Table II. Spark is used to extract schemas, process data, run queries on the EMR data and send the results back to the frontend. Spark manages data flow between EMR LH and the web interface. Healthcare professionals can access specific patient or groups' of patients data through the web interface. Our backend server connects to MongoDB and uses patients' ID to fetch video, audio, and vital signs, and link to LH systems to retrieve relevant information.

TABLE II
Description of the Tables in Delta LH

Table Name	Description
ADMISSIONS	Hospital admission associated with ICU stay.
PATIENTS	Contains information about the patients.
TRANSFERS	Tracks the movement of patients within the hospital.
DIAGNOSES_ICD	Contains ICD-9 codes which are used to identify diagnoses and procedures performed.
LABEVENTS	Contains the readings of specific test values for a patient over time from both in-hospital laboratory measurements and out-of-hospital laboratory measurements from clinics which the patient has visited.
PRESCRIPTIONS	Contains detailed information about the medications prescribed to patients during their ICU stays.
NOTEEVENTS	Contains clinical notes written by healthcare professionals.
D_LABITEMS	Contains a dictionary of definitions for laboratory measurements.

A. Experimental Setup

In our infrastructure setup, we employed a vSphere server to manage our virtualized environment efficiently. We established a VMware vSphere cluster, utilizing servers powered by industry-standard hardware such as Dell PowerEdge, running on the VMware ESXi hypervisor. We setup a vSphere cluster for the experiments with 16 CPU cores, 32 GB of RAM, and an NVIDIA V100 GPU. We have executed 12 queries for each system and recorded the end-to-end response time. For the operating system, we used Ubuntu 20.04. To ensure robust data connectivity, we utilized high-speed network connections used by the university standards to ensure reliable, secure and low-latency communication between the frontend and backend infrastructure.

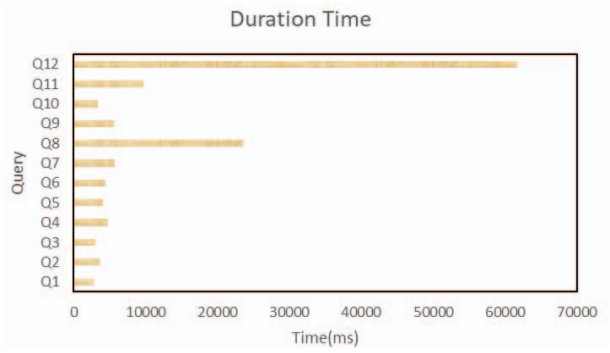


Fig. 4. Comparative review of response times of the queries executed on Delta Lakehouse.

VI. Results

Fig. 1 show the user login screen and the vital signs displayed on the screen which have values stored and retrieved from the user database implemented using MongoDB (see I,3). For login using face authentication and vital signs, users' face and/or finger videos are captured and stored in the MongoDB and used in validating the user identity and computing the vital signs. Users' emotions detected from the video and text data are also stored in the user databases and used to generate compassionate dialogues for the conversation. In this pilot

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TABLE III
QUERIES EXECUTED ON MIMIC III USING DELTA LH

No.	Query
Q1	Get the information of a patient with a specified name
Q2	Get a list of all admission types along with the count of visits for each patient
Q3	Get the information about female patients who died in a specific year

Q4	Get the information about the female patients that got transferred from one care unit to another care unit
	`SELECT Distinct P.ROW_ID,P.GENDER,T.EVENTTYPE,T.PREV_CAREUNIT,T.CURR_CAREUNIT FROM PATIENTS As P`
	`LEFT JOIN TRANSFERS AS T on P.ROW_ID=T.ROW_ID`
	`WHERE P.GENDER='F' and T.EVENTTYPE='transfer'`
	`ORDER BY P.ROW_ID`

Q5	Count the number of female patients who visited the hospital under the admission type 'NEWBORN'
Q6	Get the information about the male patients who visited the hospital
Q7	Get the information about the patients who were discharged under the admission type 'Emergency'
Q8	Get all the notes and other information about the patients who were admitted to the emergency department and later transferred to the current unit
Q9	Get all the patient admission information and care units were admitted with a possible diagnosis of Tuberculous
Q10	Get the information on different diseases and record the number of patients for each disease
Q11	Get all the prescriptions for patients diagnosed with diabetes

Q12	Get the laboratory test results, notes, and other information about patients who came to the emergency department with Diabetes related problems
	`SELECT Distinct P.ROW_ID,P.GENDER,A.ADMISSION_TYPE,LE.VALUE,`
	`LE.VALUENUM,LE.VALUEUOM,LE.FLAG,DD.LONG_TITLE,Note.TEXT,LI.CATEGORY, LI.LABEL`
	`FROM PATIENTS As P`
	`LEFT JOIN ADMISSIONS AS A on P.ROW_ID=A.ROW_ID`
	`LEFT JOIN LABEVENTS AS LE on P.ROW_ID=LE.ROW_ID`
	`LEFT JOIN D_ICD_DIAGNOSES AS DD on P.ROW_ID=DD.ROW_ID`
	`LEFT JOIN NOTEEVENTS AS Note on P.ROW_ID=Note.ROW_ID`
	`LEFT JOIN D_LABITEMS AS LI on P.ROW_ID=LI.ROW_ID`
	`WHERE A.ADMISSION_TYPE='EMERGENCY' and lower(DD.LONG_TITLE) Like lower('%Diabetes%')`
	`and lower(DD.LONG_TITLE) Like lower('%Diabetes%')`
	`ORDER BY P.ROW_ID`

prototype, we focused on 3 disease cases namely COVID, chest pain, and mental illness. Diagnosis is made based on interactions by the high-level symptoms extracted from the user by using designed dialogues and rule-based algorithms designed in consultation with doctors and nurses. The dialogue generation software module generates appropriate dialogue on the backend in response to users' statements, which are communicated back to the user through the frontend interface as shown in Fig. 1. The diagnosis module applies a rule-based approach in this pilot study to generate a diagnosis when sufficient details have been obtained from the dialogue module. The patient ranking software module takes the diagnosis from the dialogue generation and diagnosis modules, consults the EMR data for existing health issues such as high blood pressure or heart problems, and assigns a severity score based on the CTAS scale.

The experiments show that a physician or health administrator can query necessary data from the LH as needed using the Triage-Bot interface. The variation in query execution times are shown in Fig. 4, which reflects the differing natures of OLAP and OLTP queries. OLTP queries generally exhibit shorter execution times while dealing with patient records or transactions. OLAP queries such as Q12 tend to have longer execution times due to the complexity of operations involved. The relatively consistent execution times of OLTP queries (Q1-Q12) show a steady transactional system, while the longer execution time of Q12 shows a high complexity.

VII. DISCUSSION

MongoDB's flexibility allows healthcare emergency systems to store and retrieve patient data quickly, enhancing the speed and efficiency of OLTP operations. For instance, during emergency triage processes, the triage chatbot interacts with MongoDB to fetch video, audio, and text dialogue data in real-time.

The delta LH architecture complements MongoDB by providing a unified data management platform for both transactional and analytical workloads. While MongoDB caters to real-time data processing and retrieval, the Delta LH excels in handling complex analytical queries and data-driven decision-making. By combining the agility of a NoSQL database such as MongoDB with the analytical power of data lakehouse such as Delta LH, emergency healthcare systems can perform efficient data management and provide data-driven insights for decision-making.

VIII. CONCLUSION

We propose an effective and functional architecture for implementing a medical Triage-Bot supporting voice, video, and chat communications to provide a complementary medical triage service. It can reduce the crowd at the hospital emergency department, do a severity ranking based on health conditions and EMR data, and collect information for the triage nurse to advise about the next action on the patient's part. By reducing hospital wait times and letting people avail of the service from a home setting, Triage-Bot can improve the well-being of patients and medical professionals. Deployment of such a complex multifunctional system requires a robust infrastructure. In this paper, we present the infrastructure of our Triage-Bot system which supports all the frontend functionality of the bot and also allows deeper analysis of the data for diagnosis and administrative purposes. We presented the database schema for the bot implementation and

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demonstrated the advanced analytical capabilities of the data LH with screenshots showcasing the implemented features.

The bot is a complex system which can be improved in many different ways in the future. On the infrastructure perspective, we are working on deploying the bot on a public cloud and working on the security aspects of user authentication, and secure multiparty computation of the medical data. The data communication framework and transactions can be improved to support a variety of devices on the frontend. Each software module can be improved to support more disease cases for diagnosis, ranking, and dialogue generation. If real data can be obtained from a hospital triage setting, the system can be further improved using machine learning models.

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